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The Editor-in-Chief
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Dear Editor,

We are pleased to submit our manuscript, “**Carbon Footprint and Carbon-Aware Selection of Document-Oriented, Relational, and Hybrid Object-Relational Storage for Image-Based Time-Series Workloads in Green IoT**,” for consideration as a research article in *Environment Systems and Decisions*. Smart buildings and monitoring systems now keep streams of time-stamped images (for example, from cameras and sensors) in databases that run around the clock, so how those images are stored is an environmental choice as much as a technical one—yet its carbon cost has not been measured.

Using the same hardware and direct energy measurement throughout, we compared the carbon footprint of three common storage approaches: a document database (MongoDB), a relational database (PostgreSQL), and a hybrid that keeps only small descriptive records in the database while placing each image in a separate file store. Testing image qualities from low resolution up to very high resolution (6K), we found that no single option is best everywhere: the document database is the lowest-carbon choice for ordinary image sizes, the hybrid becomes the lowest-carbon choice for large images (roughly halving the emissions of storing images inside the database), and the relational database sits consistently in between. At the largest size, the document database cannot store the images at all, owing to a built-in limit on how large a single record may be. We turn these measurements into a practical, carbon-aware guide for choosing an approach, and we test how it holds up across countries with cleaner or dirtier electricity, account for the manufacturing (“embodied”) carbon of the extra storage some options consume, and project the yearly emissions of a realistic camera fleet.

The work suits *Environment Systems and Decisions*: it treats an everyday infrastructure choice as an environmental decision, combines direct measurement with both operational and manufacturing carbon, considers how the answer depends on the local electricity grid, and offers actionable guidance for sustainable building monitoring rather than a purely technical benchmark. We also relate the findings to several UN Sustainable Development Goals.

In the interest of transparency, this manuscript consolidates and substantially extends two of our earlier conference papers, each of which separately compared only two of the three approaches. Those two papers have been *accepted for presentation* at the ATIGB conference (<https://atigb.ute.udn.vn/>; University of Technology and Education – The University of Danang, Vietnam) and are awaiting presentation; we provide them as related/supporting files for your reference. The present work goes well beyond them: it measures the energy directly (rather than estimating it from an assumed, fixed power draw), brings all three approaches together under one common test, extends the range to the largest image size (where the document-database limit appears), and adds the carbon-aware decision guide and the environmental analyses that are its central contribution.

This journal manuscript is original and is not under consideration elsewhere; aside from the two conference papers disclosed above, which it extends, its results have not been published. All authors have read and approved it and declare no competing interests. To support reproducibility, the benchmark code, configuration, and raw measurements will be released in a public repository with a permanent identifier upon acceptance; for double-anonymous review, an anonymized copy (containing no author-identifying information) is available at this anonymized Zenodo record. Thank you for considering our submission.

Sincerely,

Alfa Yohannis, on behalf of the authors

(Alfa Yohannis, Universitas Pradita; Alexander Waworuntu, Universitas Multimedia Nusantara)